SHRI VIDHYABHARATHI MATRIC HIGHER SECONDARY SCHOOL

SAKKARAMPALAYAM, ELACHIPALAYAM. NAMAKKAL (Dt)

PUBLIC EXAMINATION – MARCH - 2025 (QUESTION ANALYSIS)

	1 MARK		2 MARK		3 MARK		5 MARK		TOTAL		GRAN
	B.B	INT	B.B	INT	B.B	INT	B.B	INT	B.B	INT	T TOT
1	2	-	Ex:1.20	-	1	-	1	-	12	-	12
2	1	-	-	N.P-8©	1	-	1	-	09	02	11
3	-	1	-	1	1	-	1	-	08	03	11
4	1	-	1	-	-	1	1(3)+ Ex:4.14 (2)	-	08	03	11
5	1	-	-	-	1	-	1	-	09	-	09
6	1		1	-	-	-	1	-	08	-	08
7	2	1	-	1	1	-	1(2) + Ex:7.18 (3)	-	10	03	13
8	1	-	1	-	-	Ex:8.2 (T)	1(3) + 1(2)	-	08	03	11
9	1	-	1		Ex: 9.2©	-	1	-	11	-	11
10	1	1	-	1	1	-	1	-	09	03	12
11	1	-	-	-	-	-	-	-	01	-	01
	12	3	10	8	21	6	50	2	93	17	110

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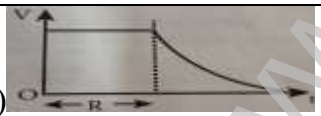
SAKKARAMPALAYAM, AGARAM (PO), ELACHIPALAYAM
TIRUCHENGODE (TK), NAMAKKAL (DT) PIN-637202
Cell: 99655-31727, 94432-31727

PUBLIC EXAMINATION – MARCH– 2025

TENTATIVE ANSWER KEY

XII -PHYSICS **TOTAL MARKS : 70**

Q.N	PART – I		MARKS
	TYPE – A	TYPE – B	15X1=15
1.	d) $\bar{A} + B + \bar{C}$	c) 25cm	1
2.	d) 0.5	c) polarisation	1
3.	c) 2Å	d) 0.5	1
4.	b) Straight line	c) 2Å	1
5.	d) $[L^2T^{-2}]$	d)	1
6.	a) sports	d) $\sqrt{\frac{6hv_0}{m}}$	1
7.	d) 8mC	a) sports	1
8.	b) 30^0	d) 8mC	1
9.	d) $\frac{N_0}{\sqrt{2}}$	b) 30^0	1
10.	c) 25cm	d) 500Å	1
11.	d) 500Å	d) All the above	1
12.	d) 	d) $[L^2T^{-2}]$	1
13.	d) $\sqrt{\frac{6hv_0}{m}}$	b) Straight line	1
14.	d) All the above	d) $\frac{N_0}{\sqrt{2}}$	1
15.	c) polarisation	d) $\bar{A} + B + \bar{C}$	1
	PART – II		
16.	$C = \frac{\varepsilon_0 A}{d}$ $= \frac{8.854\times10^{-12}\times5\times10^{-2}\times5\times10^{-2}}{10^{-3}}$ $= 221.2\times10^{-13}F$ (or) $= 22.12$ pF		1 ½ ½

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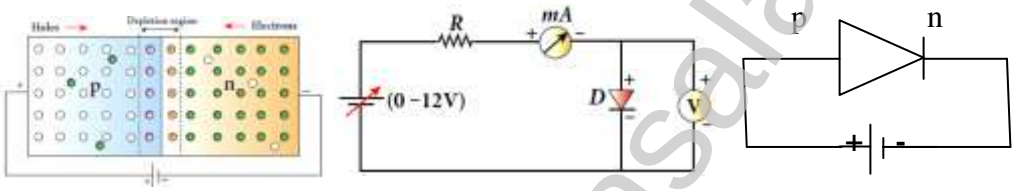
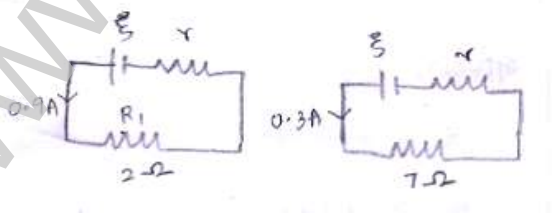
17.	**Lenz's law.** ❖ Lenz's law states that the direction of the induced current is such that it always opposes the cause responsible for its production. $\varepsilon = -N\frac{d\varphi}{dt}$ (or) $-\varepsilon = \frac{d\varphi}{dt}$ **(formula only award 1 mark)**	2
18.	**Limitations of cyclotron:** ❖ the speed of the ion is limited ❖ electron cannot be accelerated ❖ uncharged particles cannot be accelerated	2
19.	**Mass defect:** ❖ The experimental mass of any atomic nucleus is less than the total mass of its individual constituents. This difference in mass is called mass defect. $\Delta m = (Zm_p + Nm_n) - M$ **(formula only award 1 mark)**	2
20.	**Optical path:** ❖ Optical path of a medium is defined as the distance d' light travels in vacuum in the same time it travels a distance d in the medium.	2
21.	❖ infinite - plane (or) straight wave front. ❖ point source - spherical wave fronts. ❖ line source - cylindrical wave fronts. **(any two points award full marks)**	2
22.	 **(any one of the diagram)**	2
23.	**Stopping potential:** ❖ Stopping potential is that value of the negative (retarding) potential given to the collecting electrode *A* which is **just sufficient to stop the most energetic photoelectrons emitted and make the photocurrent zero.** **$1/2mv^2_{max} = eV_0$ (if formula only award 1 mark)**	2
24.	$\varepsilon = I(R + r)$ $\varepsilon_1 = I_1(R_1 + r)$ $\varepsilon_2 = I_2(R_2 + r)$  $I_1 (R_1 + r) = I_2 (R_2 + r)$ $0.9 (2 + r) = 0.3 (7 + r)$ $1.8 + 0.9 r = 2.1 + 0.3r$ $0.9 r - 0.3 r = 2.1 - 1.8$ $0.6 r = 0.3$ $r = \frac{0.3}{0.6} = 0.5\ \Omega$ (or) **(any suitable methods)**	1 ½ ½

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	PART – III	
25.	**Kirchhoff's First rule (Current rule):** ❖ It states that the algebraic sum of the currents at any junction of a circuit is zero. **Kirchhoff's Second rule (Voltage rule or Loop rule):** ❖ It states that in a closed circuit the algebraic sum of the products of the current and resistance of each part of the circuit is equal to the total emf included in the circuit.	1½ 1½
26.	$E = h\nu = \frac{hc}{\lambda} = \frac{hc}{\lambda_e}$ (eV) $= \frac{6.626\times10^{-34}\times3\times10^{8}}{300\times10^{-9}\times1.6\times10^{-19}}$ E = 4.14 eV, φ_0 **= 3.20eV** **E >φ_0 so photoelectrons are Emitted.**	1 1 ½ ½
27.	**Magnetic Lorentz force:** ❖ When an electric charge q is moving with velocity $\vec{v}$ in the magnetic field $\vec{B}$, it experiences a force, called magnetic force $\vec{F_m}$. Lorentz deduced the force experienced by a moving charge in the magnetic field $\vec{F_m}$. $\vec{F}_m = q(\vec{v}\times\vec{B})$ In magnitude, $F_m = qvB\sin\theta$ The properties of lorrentz force are 1. $\vec{F_m}$ is directly proportional to the magnetic field $\vec{B}$ 2. $\vec{F_m}$ is directly proportional to the velocity $\vec{v}$ of the moving charge 3. $\vec{F_m}$ is directly proportional to sine of the angle between the velocity and magnetic field 4. $\vec{F_m}$ is directly proportional to the magnitude of the charge q 5. The direction of $\vec{F_m}$ is always perpendicular to $\vec{v}$ and $\vec{B}$ as $\vec{F_m}$ is the cross product of $\vec{v}$ and $\vec{B}$ 6. The direction of $\vec{F_m}$ on negative charge is opposite to the direction of $\vec{F_m}$ on positive charge provided other factors are identical. 7. If velocity $\vec{v}$ of the charge q is along magnetic field $\vec{B}$ then $\vec{F_m}$ is zero **(any 4 properties)**	1 1 1
28.	**Advantages and disadvantages of AC system over DC system.** **Advantages:** ❖ The generation of AC is cheaper than that of DC. ❖ When AC is supplied at higher voltages, the transmission losses are small compared to DC transmission. ❖ AC can easily be converted into DC with the help of rectifiers. **(any 2 (or) 1 points)** **Disadvantages:** ❖ Alternating voltages cannot be used for certain applications e.g. charging of batteries, electroplating, electric traction etc. ❖ At high voltages, it is more dangerous to work with AC than DC. **(any 2 (or) 1 points)**	2(or)1 1(or)2

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29.	**Properties of Electromagnetic waves:** 1. Electromagnetic waves are produced by any accelerated charge. 2. Electromagnetic waves do not require any medium for propagation. So electromagnetic wave is a non-mechanical wave. 3. Electromagnetic waves are transverse in nature. This means that the oscillating electric field vector, oscillating magnetic field vector and propagation vector (gives direction of propagation) are mutually perpendicular to each other. The electric and magnetic fields are in the y and z directions respectively and the direction of propagation is along x direction. 4. Electromagnetic waves travel with speed which is equal to the speed of light in vacuum or free space, $c = \frac{1}{\sqrt{\varepsilon_o \mu_o}} = 3\times10^8 \, m s^{-1}$ where ε_o is the permittivity of free space or vacuum and μ_o is the permeability of free space or vacuum. 5. In a medium with permittivity ε and permeability μ, the speed of electromagnetic wave is less than speed in free space or vacuum, that is, v < c. In a medium of refractive index, $n = \frac{c}{v} = \frac{\frac{1}{\sqrt{\varepsilon_o \mu_o}}}{\frac{1}{\sqrt{\varepsilon \mu}}} \Rightarrow n = \sqrt{\varepsilon_r \mu_r}$ where ε_r is the relative permittivity of the medium (also known as dielectric constant) and μ_r is the relative permeability of the medium. 6. Electromagnetic waves are not deflected by electric field or magnetic field. 7. Electromagnetic waves can show interference, diffraction and can also be polarized. **8.** Like other waves, electromagnetic waves also carry energy, linear momentum and angular momentum. **(any 6 points)**	3
30	**Relation between phase difference and path difference.** ❖ Phase is the angular position of a vibration. one wavelength λ corresponds to a phase of 2π ❖ A path difference δ corresponds to a phase difference ϕ as given by the equation, $\delta = \frac{\lambda}{2\pi}\times\phi$ (or) $\phi = \frac{2\pi}{\lambda}\times\delta$	1 1 1

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31. **Difference between Coulomb and Gravitational force:**

- The gravitational force between two masses is always attractive but Coulomb force between two charges can be attractive or repulsive, depending on the nature of charges.
- The value of the gravitational constant $G = 6.67 \times 10^{-11}$ N m^2 kg^{-2}. The value of the constant k in Coulomb law is $k = 9 \times 10^9$ N m^2 C^{-2}. Since k is much more greater than G, the electrostatic force is always greater in magnitude than gravitational force for smaller size objects.
- The gravitational force between two masses is independent of the medium. For example, if 1 kg of two masses are kept in air or inside water, the gravitational force between two masses remains the same. But the electrostatic force between the two charges depends on nature of the medium in which the two charges are kept at rest.
- The gravitational force between two point masses is the same whether two masses are at rest or in motion. If the charges are in motion, yet another force (Lorentz force) comes into play in addition to coulomb force.
- The force on a charge q_1 exerted by a point charge q_2 is given by

$$\vec{F}_{12} = \frac{1}{4\pi\varepsilon_\circ}\frac{q_1 q_2}{r^2}\hat{r}_{21}$$

Here $\hat{r}_{21}$ is the unit vector from charge q_2 to q_1.

But $\hat{r}_{21} = -\hat{r}_{12}$,

$$\vec{F}_{12} = \frac{1}{4\pi\varepsilon_\circ}\frac{q_1 q_2}{r^2}(-\hat{r}_{12}) = -\frac{1}{4\pi\varepsilon_\circ}\frac{q_1 q_2}{r^2}(\hat{r}_{12})$$

(or) $\vec{F}_{12} = -\vec{F}_{21}$

Therefore, the electrostatic force obeys Newton's third law.

- The expression for Coulomb force is true only for point charges. But the point charge is an ideal concept. However we can apply Coulomb's law for two charged objects whose sizes are very much smaller than the distance between them.

(any 3 comparisons) 3

32.

Intrinsic semiconductor	**Extrinsic semiconductor**
semiconductor without impurity (pure form)	semiconductor with impurity (impure form)
Has equal number of charge carriers concentration	In N-type electrons are majority charge carriers In P-type holes are majority charge carriers
Electrical conductivity is less	Electrical conductivity is high
eg., Germanium and silicon.	eg., silicon doped with boron, Germanium doped with arsenic N- type, P –type.

(any three points) 3

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33.

(i) Angular momentum

$l = n\hbar = \frac{nh}{2\pi}$ ½

$= \frac{5\times6.6\times10^{-34}}{2\times3.14}$ ½

$=5.25\times10^{-34}\ kgm^2s^{-1}$ ½

(ii) velocity is given by

$v = \frac{l}{mr}$ ½

$= \frac{5.25\times10^{-34}}{(9.1\times10^{-31})(13.25\times10^{-10})}$ ½

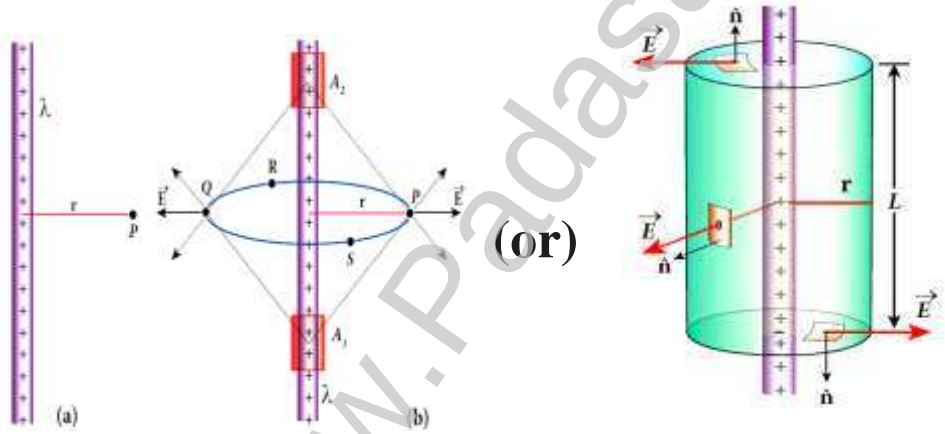
$= 4.4\times10^{5}\ ms^{-1}$ ½

PART – IV

34. a)

Expression for electric field due to an infinitely long charged wire.

- Consider an infinitely long straight wire having uniform linear charge density λ. Let P be a point located at a perpendicular distance r from the wire.
- The electric field at the point P can be found using Gauss law. We choose two small charge elements A_1 and A_2 on the wire which are at equal distances from the point P.
- The resultant electric field due to these two charge elements points radially away from the charged wire and the magnitude of electric field is same at all points on the circle of radius r. 1
- The charged wire possesses a cylindrical symmetry.



1

- Let us choose a cylindrical Gaussian surface of radius r and length L as shown in the Figure.

  The total electric flux in this closed surface is

$$\Phi_E = \oint \vec{E}\cdot d\vec{A}$$

$$= \int_{\substack{Curved\\surface}} \vec{E}\cdot d\vec{A} + \int_{\substack{top\\surface}} \vec{E}\cdot d\vec{A} + \int_{\substack{bottom\\surface}} \vec{E}\cdot d\vec{A} \quad \text{..... (1)}$$

1

- It is seen from Figure that for the curved surface, $\vec{E}$ is parallel to $\vec{A}$ and $\vec{E}.\overrightarrow{dA} = EdA$.
- For the top and bottom surfaces, $\vec{E}$ is perpendicular to $\vec{A}$ and, $\vec{E}.\overrightarrow{dA} = 0$ Substituting these values in the equation(1) and applying Gauss law to the cylindrical surface, we have

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$$\Phi_E = \int_{\substack{Curved\\surface}} E\,dA = \frac{Q_{encl}}{\varepsilon_\circ} \quad \ldots\ldots\ldots\ldots\ldots (2)$$ ½

❖ Since the magnitude of the electric field for the entire curved surface is constant, E is taken out of the integration and $Q_{enclosed}$ is given by

$$E \int_{\substack{Curved\\surface}} dA = \frac{\lambda L}{\varepsilon_\circ}$$

Here $\int_{\substack{Curved\\surfa}} dA =$ total area of the curved surface $= 2\pi rL$. ½

$$E \cdot 2\pi rL = \frac{\lambda L}{\varepsilon_\circ}$$

$$E = \frac{1}{2\pi\varepsilon_\circ}\frac{\lambda}{r}$$ 1

In vector form $\vec{E} = \frac{1}{2\pi\varepsilon_\circ}\frac{\lambda}{r}\hat{r}$

❖ The electric field due to the infinite charged wire depends on 1/r rather than $1/r^2$ for a point charge.

or 34. (b)

De Morgan's First Theorem

❖ The first theorem states that the complement of the sum of two logical inputs is equal to the product of its complements. 1

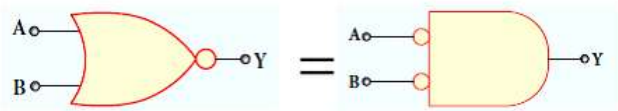
Proof

The Boolean equation for NOR gate is $Y = \overline{A+B}$

The Boolean equation for a bubbled AND gate is $Y = \overline{A}.\overline{B}$

A	B	A+B	$\overline{A+B}$	$\overline{A}$	$\overline{B}$	$\overline{A}.\overline{B}$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

1



From the above truth table, we can conclude $\overline{A+B} = \overline{A}\cdot\overline{B}$. ½

De Morgan's Second Theorem

❖ The second theorem states that the complement of the product of two inputs is equal to the sum of its complements. 1

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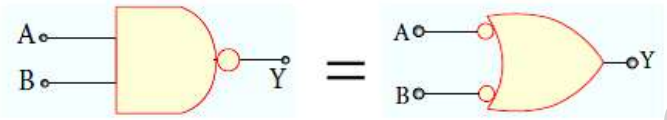
Proof

The Boolean equation for NAND gate is $Y = \overline{AB}$

The Boolean equation for bubbled OR gate is $Y = \overline{A} + \overline{B}$

A	B	A.B	$\overline{A.B}$	$\overline{A}$	$\overline{B}$	$\overline{A}+\overline{B}$
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

1



From the above truth table we can conclude $\overline{A.B} = \overline{A} + \overline{B}$

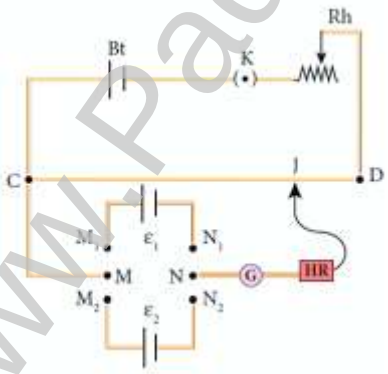
½

35. a)

Comparision of emf of two cells:

- To compare the emf of two cells, the circuit connections are made as shown in Figure.
- Potentiometer wire CD is connected to a battery *Bt* and a key K in series.
- This is the primary circuit. The end C of the wire is connected to the terminal M of a DPDT (Double Pole Double Throw) switch and the other terminal N is connected to a jockey through a galvanometer G and a high resistance HR.
- The cells whose emf ε_1 and ε_2 to be compared are connected to the terminals M_1,N_1 and M_2,N_2 of the DPDT switch.
- The positive terminals of *Bt*, ε_1 and ε_2 should be connected to the same end C.

1



1

- The DPDT switch is pressed towards M_1, N_1 so that cell ε_1 is included in the secondary circuit and the balancing length l_1 is found by adjusting the jockey for zero deflection.
- Then the second cell ε_2 is included in the circuit and the balancing length l_2 is determined.
- Let *r* be the resistance per unit length of the potentiometer wire and *I* be the current flowing through the wire.

we have $\varepsilon_1 = Irl_1$

1

$\varepsilon_2 = Irl_2$

1

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$$\frac{\varepsilon_1}{\varepsilon_2} = \frac{l_1}{l_2}$$ 1

- By including a rheostat (R_h) in the primary circuit, the experiment can be repeated several times by changing the current flowing through it.

35. (OR) b)

Fizeau's method to determine speed of light:

Apparatus:

- The apparatus used by Fizeau for determining speed of light in air medium. The light from the source *S* was first allowed to fall on a partially silvered glass plate *G* kept at an angle of 45°to the incident light.
- The light then was allowed to pass through a rotating toothed-wheel with *N* teeth and *N* cuts of equal widths whose speed of rotation could be varied through an external mechanism (not shown in the Figure). 1
- The light passing through one cutin the wheel will get reflected by a mirror *M* kept at a long distance *d*, about 8 *km* from the toothed wheel.
- If the toothed wheel was not rotating, the reflected light from the mirror would again pass through the same cut and reach the eyes of the observer through the partially silvered glass plate.

Working:

- The angular speed of rotation of the toothed wheel was increased from zero to a value ω until light passing through one cut would completely be blocked by the adjacent tooth.
- This is ensured by the disappearance of light while looking through the partially silvered glass plate.

Expression for speed of light:

- The speed of light in air *v* is equal to the ratio of the distance the light travelled from the toothed wheel to the mirror and back 2*d* to the time taken *t*.

$$v = \frac{2d}{t}$$

- The distance *d* is a known value from the arrangement. The time taken *t* for the light to travel the distance to and fro is calculated from the angular speed ω of the toothed wheel. 1
- The angular speed ω of the toothed wheel when the light disappeared for the first time is,

$$\omega = \frac{\theta}{t}$$

- Here, θ is the angle between the tooth and the slot which is rotated by the toothed wheel within that time *t*.

$$\theta = \frac{\text{total angle of the circle in radian}}{\text{number of teeth+number of cuts}}$$ ½

$$\theta = \frac{2\pi}{2N} = \frac{\pi}{N}$$

Substituting for θ in the equation for ω,

$$\omega = \frac{\pi / N}{t} = \frac{\pi}{Nt}$$

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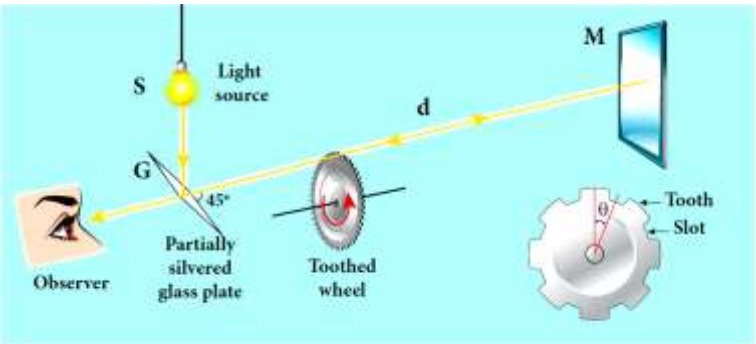
$$t = \frac{\pi}{N\omega}$$ ½



1

Substituting t in equation

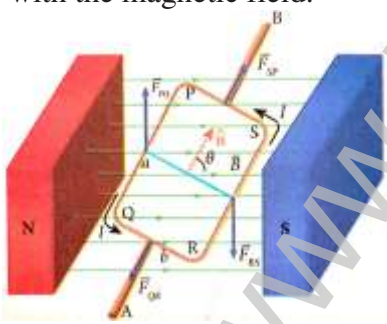
$$v = \frac{2d}{\pi / N\omega}$$

$$v = \frac{2dN\omega}{\pi}$$ ½

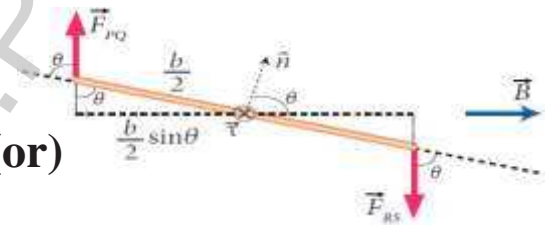
❖ Fizeau had some difficulty to visually estimate the minimum intensity of the light when blocked by the adjacent tooth, and his value for speed of light was very close to the actual value. Later on, with the same idea of Fizeau and with much sophisticated instruments, the speed of light in air was determined as, $v = 2.99792 \times 10^8$ m s^{-1}. ½

36. a) **The expression for the torque on a current carrying coil in a magnetic field.**

❖ Consider a rectangular loop PQRS carrying current I is placed in a uniform magnetic field $\vec{B}$. Let a and b be the length and breadth of rectangular loop respectively. The unit vector $\hat{n}$ normal to the plane of the loop makes an angle θ with the magnetic field.



(or)



(any one diagram) 1

❖ The magnitude of the magnetic force acting on the current carrying arm PQ is

$F_{PQ} = IaB \sin(\pi/2) = IaB$

The direction of force is upwards.

❖ The magnitude of the magnetic force acting on the current carrying arm QR is

$F_{QR} = IaB \sin(\pi/2 - \theta) = IaB\cos\theta$

❖ The magnitude of the magnetic force acting on the current carrying arm RS is

$F_{RS} = IaB \sin(\pi/2) = IaB$

The direction of force is downwards.

❖ The magnitude of the magnetic force acting on the current carrying arm SP is

$F_{SP} = IaB \sin(\pi/2 + \theta) = IaB\cos\theta$ **(upto this)** 1

Since the forces F_{QR} and F_{SP} are equal, opposite and collinear, they cancel each

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other. But the forces F_{PQ} and F_{RS}, which are equal in magntitude and opposite in direction are not acting along the same straight line. Therefore, F_{PQ} and F_{RS} constitute a couple which exerts a torque on the loop.

❖ The magnitude of torque acting on the arm PQ about AB is

$$\tau_{PQ} = \left(\frac{b}{2}\sin\theta\right) IaB$$

and it points in the direction of AB

❖ The magnitude of torque acting on the arm RS about AB is

$$\tau_{RS} = \left(\frac{b}{2}\sin\theta\right) IaB$$

and it points in the same direction

❖ The total torque acting on the entire loop about an axis AB is given by

$$\tau = \left(\frac{b}{2}\sin\theta\right) F_{PQ} + \left(\frac{b}{2}\sin\theta\right) F_{RS}$$

$$= Ia(b\sin\theta)B$$ **(upto this)** 1

$$\tau = IAB\sin\theta \text{ along the direction AB}$$

❖ In vector form

$$\vec{\tau} = \left(I\vec{A}\right) \times \vec{B}$$ ½

In terms of magnetic dipole moment

$$\vec{\tau} = \vec{p}_m \times \vec{B} \text{ where } \vec{p}_m = I\vec{A}$$ ½

❖ The tendency of the torque is to rotate the loop so as to align its normal vector with the direction of the magnetic field.

❖ If there are N turns in the rectangular loop, then the torque is given by 1

$$\tau = NIAB\sin\theta$$

36 (OR) b)

Photo emissive cell:

Construction

❖ It consists of an evacuated glass or quartz bulb in which two metallic electrodes – that is, a cathode and an anode are fixed as shown in Figure

❖ The cathode *C* is semi-cylindrical in shape and is coated with a photo sensitive material. The anode *A* is a thin rod or wire kept along the axis of the semi-cylindrical cathode. 1

❖ A potential difference is applied between the anode and the cathode through a galvanometer *G*.

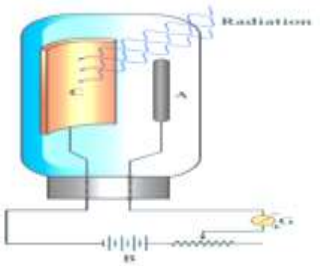
Working:

❖ When cathode is illuminated, electrons are emitted from it. These electrons are attracted by anode and hence a current is produced which is measured by the galvanometer. ½

For a given cathode, the magnitude of the current depends on

i) the intensity to incident radiation and

ii) the potential difference between anode and cathode. ½



1

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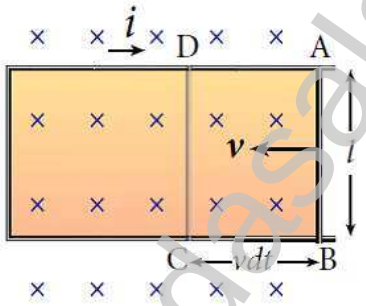
Applications of photo cells:

- Photo cells have many applications, especially as switches and sensors. Automatic lights that turn on when it gets dark use photocells, as well as street lights that switch on and off according to whether it is night or day use photocells.
- Photo cells are used for reproduction of sound in motion pictures and are used as timers to measure the speeds of athletes during a race.
- Photo cells of exposure meters in photography are used to measure the intensity of the given light and to calculate the exact time of exposure.

(any 2 points) — 2

37. (a) **(i) Induce an emf by changing the area enclosed by the coil:**

- Consider a conducting rod of length l moving with a velocity towards left on a rectangular metallic framework as shown in Figure. — ½
- The whole arrangement is placed in a uniform magnetic field whose magnetic lines are perpendicularly directed into the plane of the paper.
- As the rod moves from AB to DC in a time dt, the area enclosed by the loop and hence the magnetic flux through the loop decreases.



× → $\vec{B}$ (⊥r, inwards) — ½

- The change in magnetic flux in time dt is

$$d\Phi_B = B \times \text{ change in area}$$

$$= B \times \text{ Area ABCD}$$ — 1

$$\text{Since Area } ABCD = l(vdt)$$

$$d\Phi_B = Blvdt \quad \text{(or)}$$

$$\frac{d\Phi_B}{dt} = Blv$$ — 1

- As a result of change in flux, an emf is generated in the loop. The magnitude of the induced emf is

$$\varepsilon = \frac{d\Phi_B}{dt}$$

$$\varepsilon = Blv$$

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	This emf is called motional emf. The direction of induced current is found to be clock wise from Fleming's right hand rule. (ii) $\frac{Area}{Time}$ = Area × Frequency $\frac{dA}{dt} = 0.03 \times 20 = 0.6\ m^2\ s^{-1}$ $\varepsilon = \frac{d\varphi_B}{dt} = \frac{d}{dt}BA = B\frac{dA}{dt}$ $= \frac{0.4 \times 0.6}{1} = 0.24V$	 ½ 1 ½
37. (b)	**(i) Brewster's law :** ❖ The law states that the tangent of the polarising angle for a transparent medium is equal to its refractive index. $\tan i_p = n$ **(formula only award 1 mark)** **(ii)** $i_P = \tan^{-1}(n)$ $= \tan^{-1}(1.65)$ $= 58.8^0$ Angle of inclination = $90^0 - 58.8^0$ $= 31.2^0$	2 1 ½ ½ 1
38. (a)	**law of radioactivity:** ❖ At any instant *t*, the number of decays per unit time, called rate of decay is proportional to the number of nuclei (dN / dt) at the same instant. **(or)** $\frac{dN}{dt} \propto N$ $\frac{dN}{dt} = -\lambda N$ $dN = -\lambda N dt$ The negative sign implies that N is decreasing with time. at time $t = 0$ s, the number of nuclei present in the radioactive sample is N_0 $\frac{dN}{N} = -\lambda\, dt$ $\int_{N_0}^{N} \frac{dN}{N} = -\int_0^t \lambda\, dt$ $[\ln N]_{N_0}^{N} = -\lambda t$ $\ln\left[\frac{N}{N_0}\right] = -\lambda t$ ❖ Taking exponentials on both sides, we get $N = N_0 e^{-\lambda t}$ ❖ This Equation is called the law of radioactive decay. ❖ Here *N* denotes the number of un decayed nuclei present at any time *t* and N_0denotes the number of nuclei at initial time *t=0*.**The number of atoms is decreasing exponentially over the time**.	 1 1 1 1

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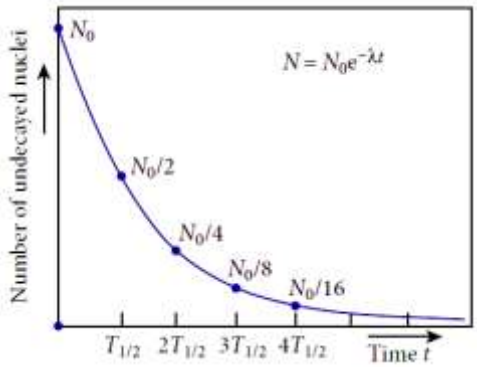
	Time taken for all the radioactive nuclei to decay will be infinite.  	1
38. (b)	**Absorption spectrum.** ❖ **When light is allowed to pass through a medium or an absorbing substance then the spectrum obtained is known as absorption spectrum.** It is the characteristic of absorbing substance. Absorption spectrum is classified into three types: **(i) Continuous absorption spectrum** ❖ When the light is passed through a medium, it is dispersed by the prism, we get continuous absorption spectrum. For instance, when we pass white light through a blue glass plate, it absorbs everything except blue. This is an example of continuous absorption spectrum. **(ii) line absorption spectrum** ❖ When light from the incandescent lamp is passed through cold gas (medium), the spectrum obtained through the dispersion due to prism is line absorption spectrum. ❖ Similarly, if the light from the carbon arc is made to pass through sodium vapour, a continuous spectrum of carbon arc with two dark lines in the yellow region of sodium vapour is obtained. **(iii) Band absorption spectrum** ❖ When the white light is passed through the iodine vapour, dark bands on continuous bright background is obtained. ❖ This type of band is also obtained when white light is passed through diluted solution of blood or chlorophyll or through certain solutions of organic and inorganic compounds. **(only names mentioned award 1 mark)**	2 1 1 1

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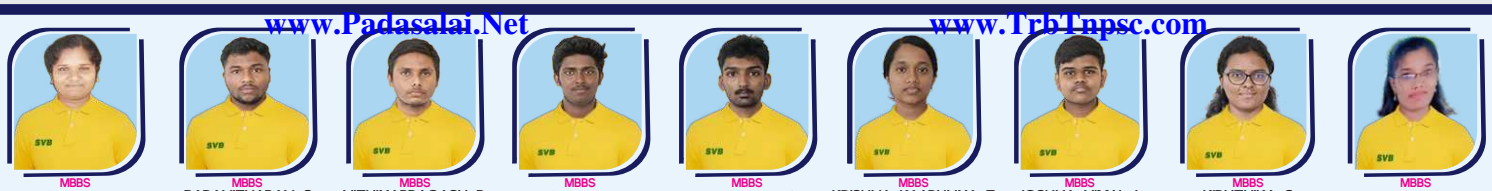

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